

[ISOM2010] INTRODUCTION TO INFORMATION SYSTEMS

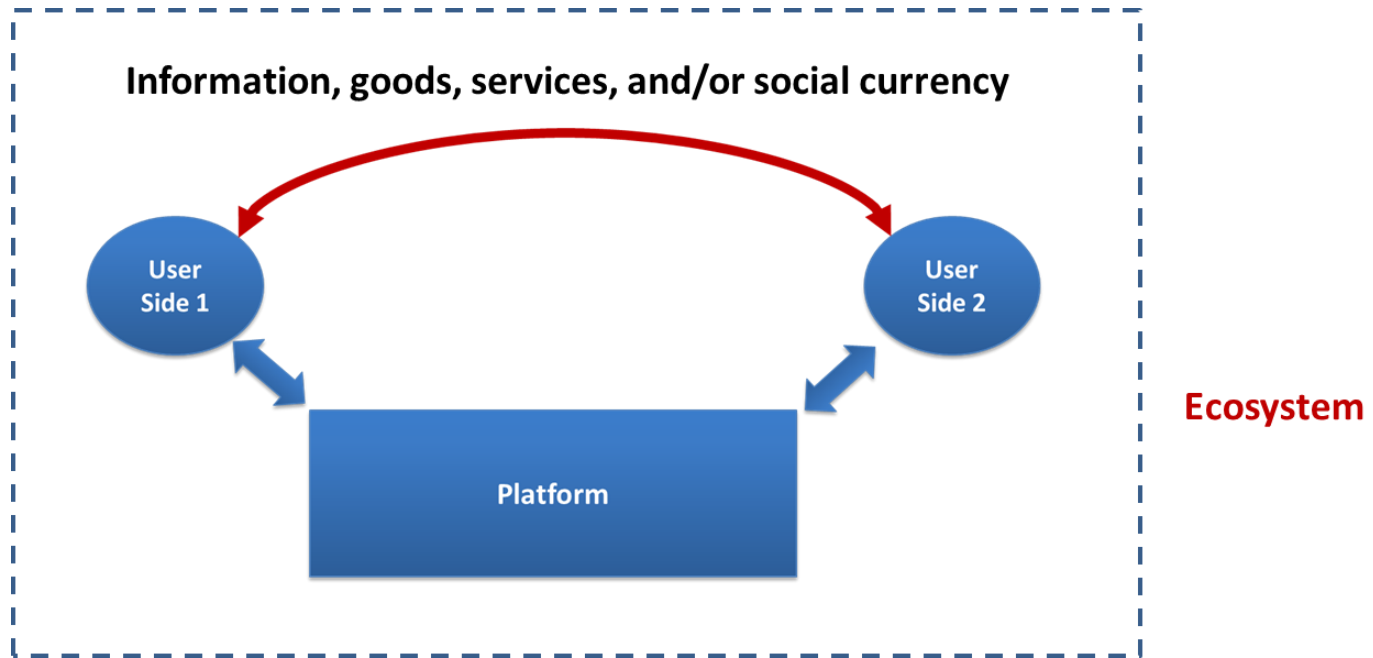
WHAT IS A DIGITAL PLATFORM?

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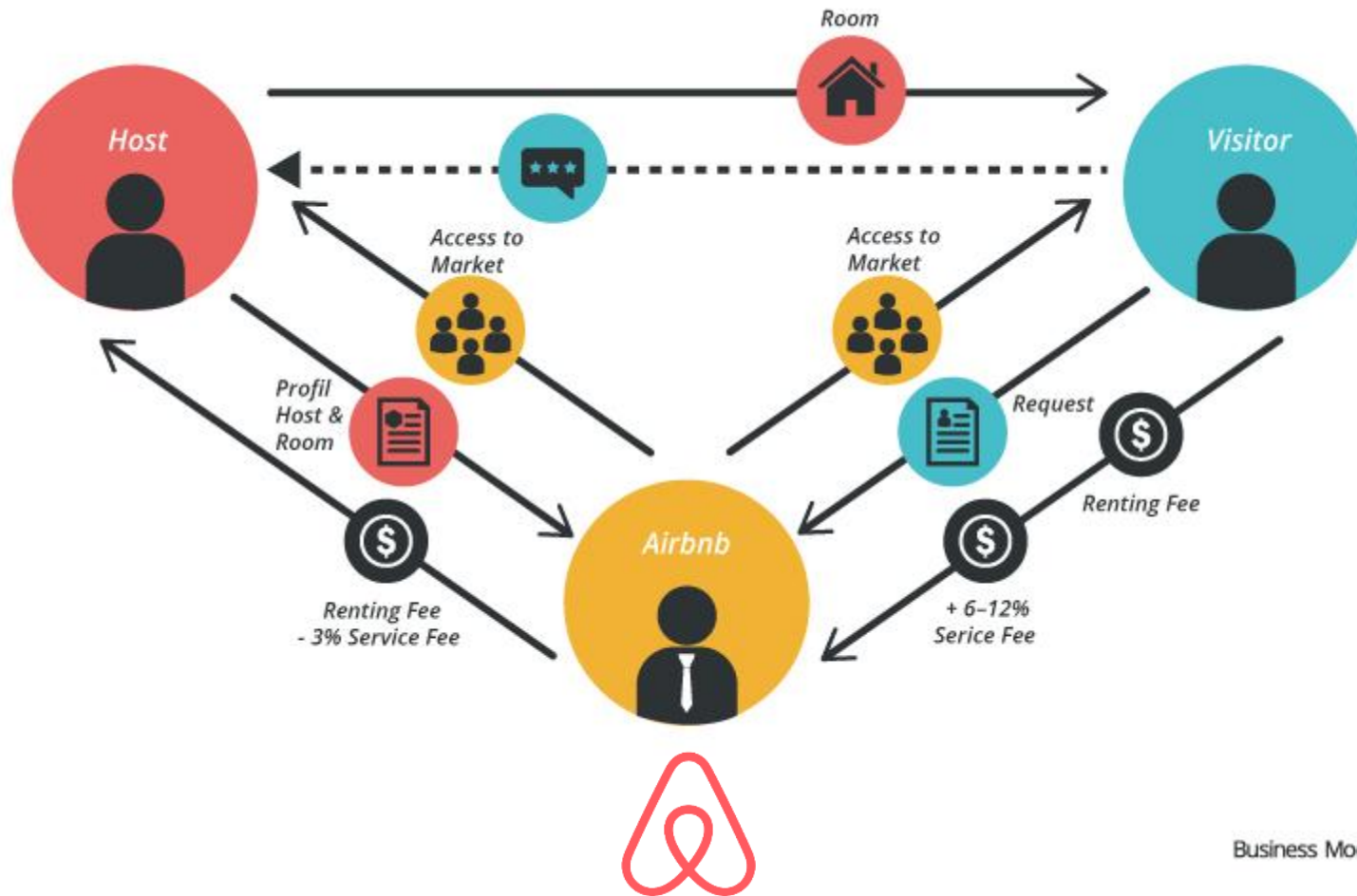
PROF. KIM, YONGSUK

What is a platform?

Platforms bring together individuals and organizations so they can interact or innovate with one another in ways not otherwise possible, with the potential for *nonlinear increase* in utility and value.



Airbnb: a two-sided platform



Business Model **Toolbox**

Source: <http://bmtoolbox.net/patterns/sharing-economy/>

Platforms bring together individuals and organizations so they can interact or innovate with one another in ways not otherwise possible, **with the potential for *nonlinear increase* in utility and value.**

Network Effects

Positive feedback loops that come from **connecting** different users and market participants to each other

The Morning Download: Facebook Earnings Reflect **Epic Network Effect**

ARTICLE

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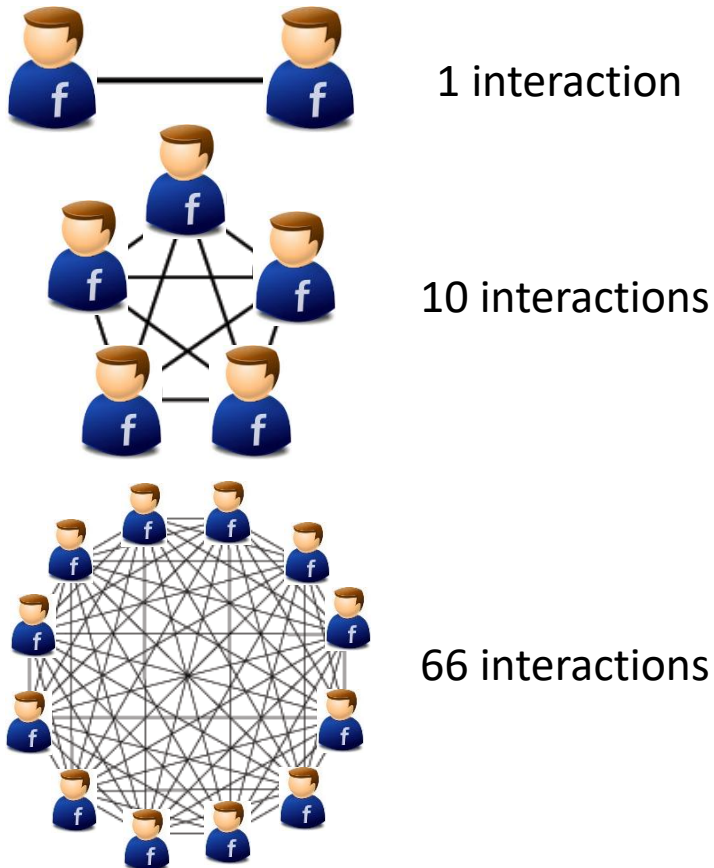
By STEVE ROSENBUSH [CONNECT](#)



Good morning. **Facebook** Inc. **FB +1.31%** posted strong third-quarter earnings by almost every possible metric, including earnings, mobile, and video. It achieved that growth even though it continued to invest in new initiatives, “including artificial intelligence, virtual reality and bringing Internet access to emerging markets,” **the WSJ’s Deepa Seetharaman reports**. It’s a perfect illustration of the winner-take-all tendency in technology, the network effect writ large. One in seven people on planet earth check Facebook every day, **Wired says**.

Network effect

- The impact that the number of users of a platform has on the value created for each other.



The value of FB increases as its number of users expands (value increases with n^2)

Same digital service but different value

Apple Pay Statistics

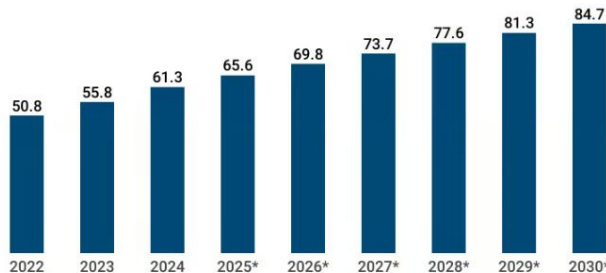
Last Updated: January 28, 2026

Highlights. Worldwide, **785 million** people use Apple Pay, making it the **#1 most popular digital wallet**.

- Projections indicate U.S. Apple Pay usership will grow from **65.6 million** in 2025 to over **84 million** by 2030.
- Apple Pay is accepted at over **90%** of U.S. retailers and available in **95 countries** worldwide.
- Consumers use Apple Pay in **14.2%** of all online payments.
- Apple Pay processed **\$8.5 trillion** in payments in 2024.
- In the U.S., Apple Pay has a **92%** market share.

American Apple Pay Users

(in millions; consumers aged 14+)



Source: eMarketer

*Projection

The KoreaTimes

Business > Banking & Finance

Two years in, Apple Pay struggles to gain foothold in Korea



Listen



By Lee Yeon-woo

Published Mar 14, 2025 4:46 pm KST | Updated Mar 17, 2025 1:33 pm KST

Card issuers say it is poisoned chalice

"There are too few places where I can use it. I just carry around physical credit cards nowadays," she said.

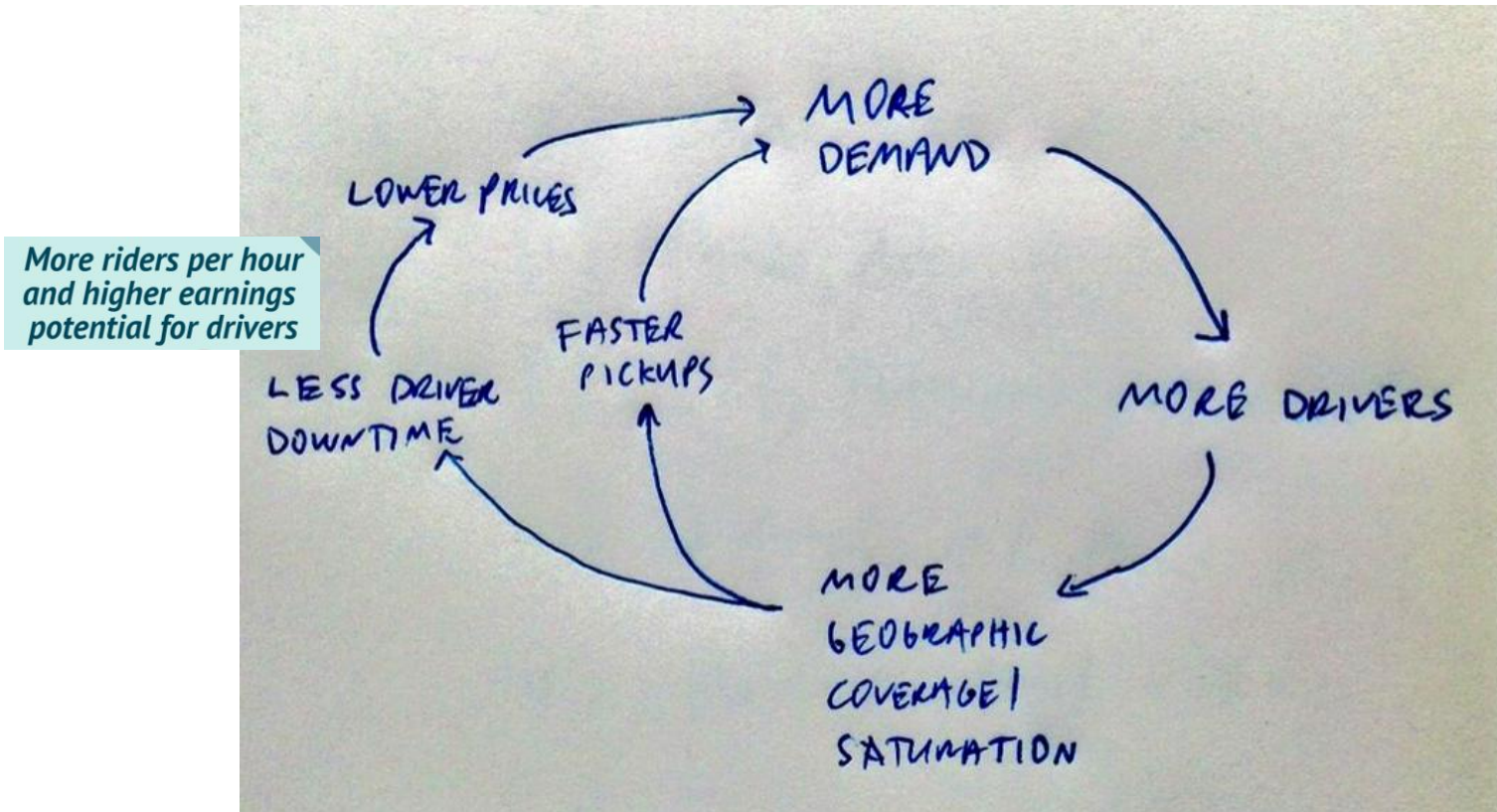
In 2026, the percentage of retailers in South Korea accepting Apple Pay remains low, estimated at approximately 10%.

<https://www.koreatimes.co.kr/business/banking-finance/20250314/two-years-in-apple-pay-struggles-to-gain-foothold-in-korea>

Powerful platforms and network effect

Having more users and content/product/service providers leads to a more efficient system. There is a significant advantage to being the largest player in any given market.

Positive Feedback loop between SCALE and SATISFACTION



This is a famous sketch drawn by David Sacks – a silicon valley VC. He drew the picture above describing how Uber's business model was based on a positive feedback loop.

Two-sided Platforms

eBay Sellers
Expedia Airlines/Hotels
Xbox Developers
American Express Merchants
Aga Khan Doctors
YouTube Videographers
AirBnb Rooms
Electric Car Charge Stations
Mechanical Turk Laborers
LinkedIn Employers
Android Developers



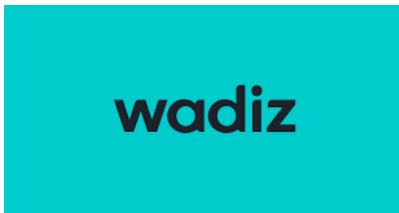
eBay Buyers
Expedia Travelers
Xbox Gamers
Amex Cardholders
Aga Khan Patients
YouTube Viewers
AirBnb Renters
Electric Car Drivers
Mechanical Turk Jobs
LinkedIn Employees
Android Users

Three-sided platform



Crowdfunding and Crowdsourcing

Crowdfunding



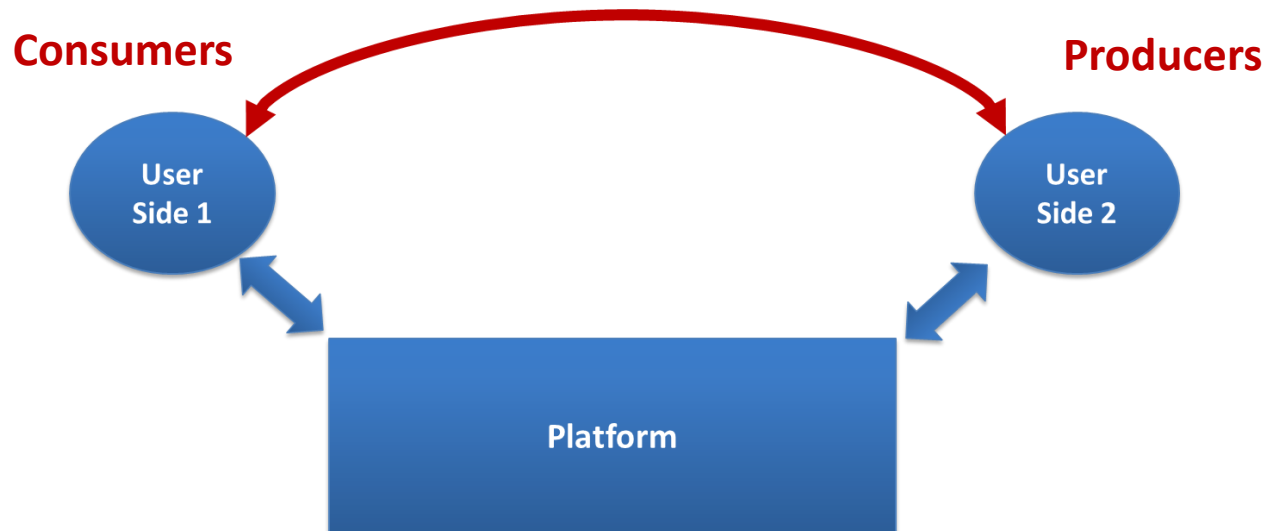
Crowdsourcing



Other multi-sided platform examples?

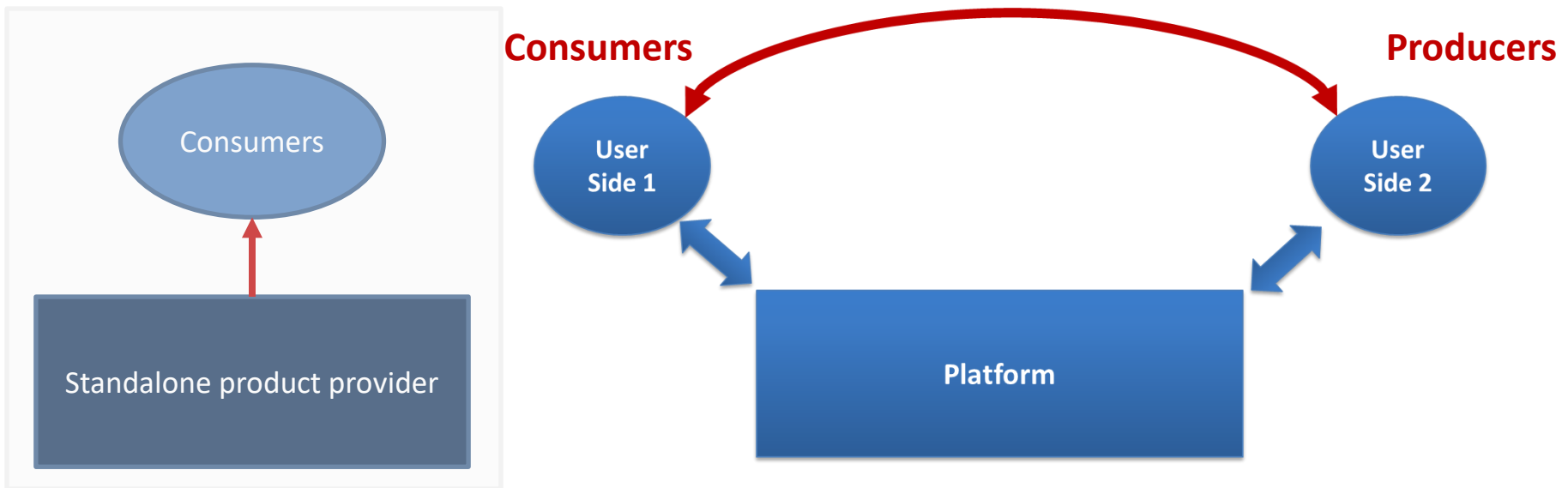
More on the concept of a platform

- A business based on enabling **value-creating interactions** between **external (3rd party) producers** and **consumers**.
- A platform provides an open, participatory infrastructure for these interactions and sets governance conditions for them.
- A platform can be hardware, software, or even a shopping mall.

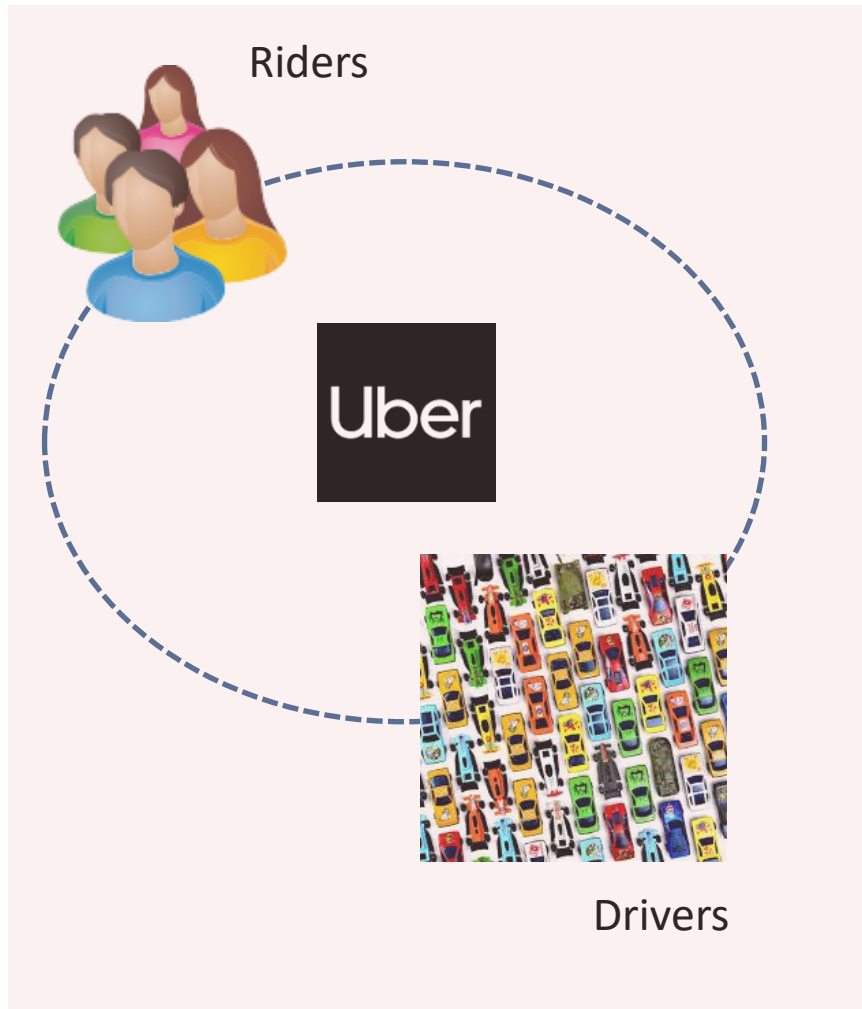


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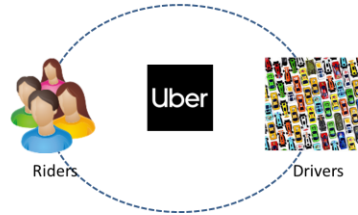


Transaction Platform vs. Innovation Platform



Games are **complementary goods** for a game console;
Gamers (consumers) value them *together*

Transaction Platform vs. Innovation Platform



	Transaction Platform	Innovation Platform
Definition	The platform serves as an intermediary for direct exchange or transactions	The platform serves as a technological foundation upon which other firms develop complementary innovations
Examples	Twitter, Instagram, Uber, LinkedIn, and FoodPanda	Nintendo, Windows, and Apple iOS
Network Effects	Come from the increasing number of participants, functions, content or services	Come from the increasing number or utility of the <u>complements</u> *
Value Creation	By facilitating the buying or selling of goods or other interaction between users	By facilitating the development of new complementary products and services

* Complements are innovations that add functionality or access to assets that make the platform increasingly useful.

Other innovation platform examples?

Standalone products vs. platforms



Based on technical quality and features (stand-alone value)



Based on **network effects** AND/OR technical quality and features

Stand-alone value vs. Network effect value

Stand-alone Value

The value of technological utility such as functionalities, simplicity, convenience, design, ease of use, risk, etc.



Network Effect Value

Network effect exists when the value of a “product” increases with the number of other users of the “product”

